

Score: / **10**

Calculus I (MATH 2200)

Spring 2020

Wednesday, 03/11/2020

Quiz 7: Section 3.8, 3.9, 3.10

Name (Print): _____

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Name (Print): _____

Section 3.8

3 pts

1. Find y' given that y satisfies the equation $\sin(xy) = x^2 + y$.

Section 3.9

3 pts

1. Find the derivative of $g(t) = 108(2^{t/12})$.

1 (bonus)

2. Find the derivative of $f(x) = x^{\sin(x)}$.

Section 3.10

2 pts

1. The function $f(x) = \sqrt{x} + x^2 + 1$ is one-to-one for $x \geq 0$ and has an inverse on that interval. Find the slope of the curve $y = f^{-1}(x)$ at the point $(3, 1)$.

2 pts

2. Find the equation of the line tangent to the graph of $g(x) = \sec^{-1}(2x)$ at the point $(1, \pi/3)$.